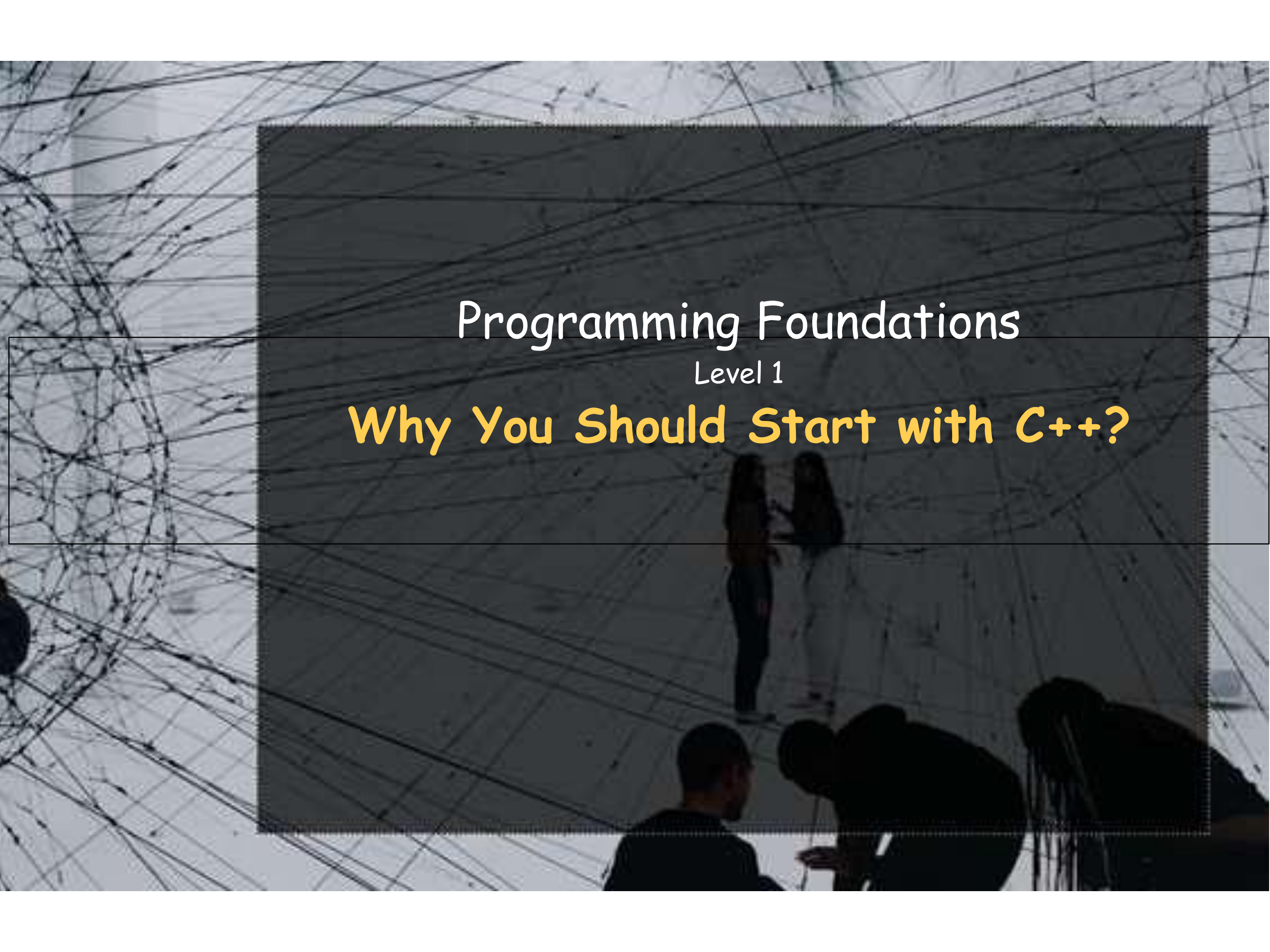




Programming Foundations Level 1

The background of the slide features a dark, moody photograph of a meeting room. In the foreground, the silhouettes of several people are visible, some standing and some sitting, engaged in discussion. In the background, a large screen displays a complex network diagram with numerous interconnected nodes and lines. The overall atmosphere is professional and collaborative.

Programming Foundations

Level 1

Why You Should Start with C++?

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing a light gray background, resembling a neural network or a web. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. A large, dark gray rectangular area is overlaid on the right side of the image, serving as a backdrop for the text.

Programming Foundations

Level 1

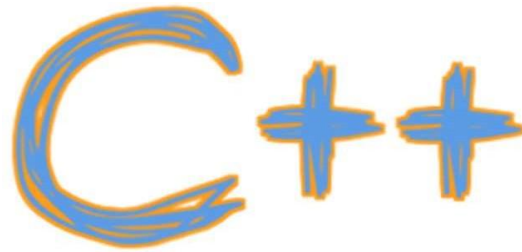
What is C++?

What/Why C++?

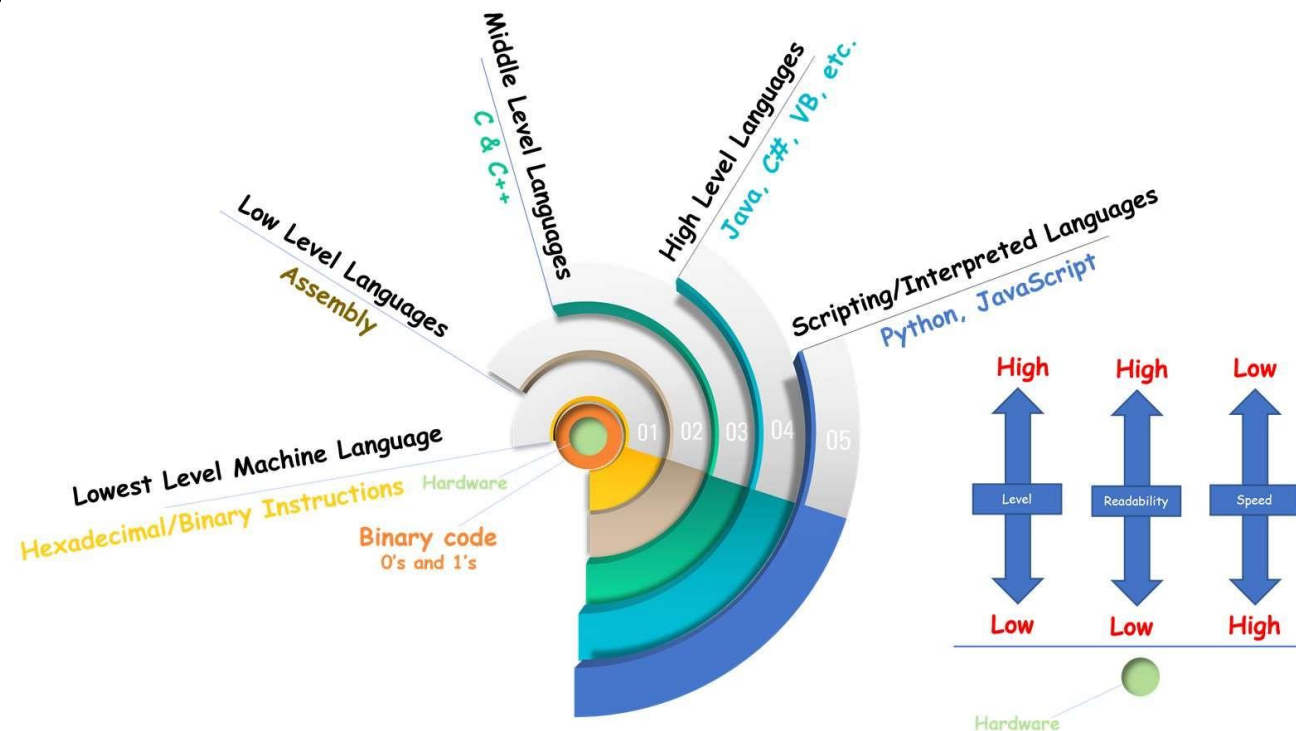


- C++ is a **cross-platform/Portable language** that can be used to create high-performance applications that can be adapted to multiple platforms
- C++ is fun and easy to learn!
- C++ is the mother of all languages.
- It Supports Procedural & OOP programming.
- It will force you learn basic programming concepts.
- If you learn it very well, you can learn any other programming language in less than a month's time.

What/Why C++?



- Both C and C++ are **considered Mid-level languages** that are close to hardware and are able to program and manipulate the underlying hardware resources.
- C and C++ have full control on computer resources such as memory, no automatic memory management.



Cross-platform/portable and used to build **high-performance** apps.

Fun and easy to learn; known as the “**mother of all languages.**”

Supports **procedural & OOP** paradigms → you learn **both styles**.

Forces core concepts; learn it well → **learn any other language in < 1 month**.

Closer to hardware:

C and C++ are **mid-level**: close to hardware, able to manipulate resources.

Full control of memory (no automatic memory management).



Programming Foundations

Level 1

What can you do with C++?

Operating systems, enterprise software, embedded systems, medical & engineering apps, databases, IDEs, compute platforms, web browsers, compilers, game engines, and desktop apps.

Real products built :

OS: Parts of Apple OS X; many **Microsoft Windows** versions built with C++ (Visual C++).

Apps: Adobe Illustrator/Photoshop core functions; Microsoft Word/Excel/PowerPoint; Amazon processing core.

Browsers / Databases / IDEs / Game Engines: highlighted as major C++ domains.

Enterprise & Embedded: flight sims, radar, cars/flights/medical machines.

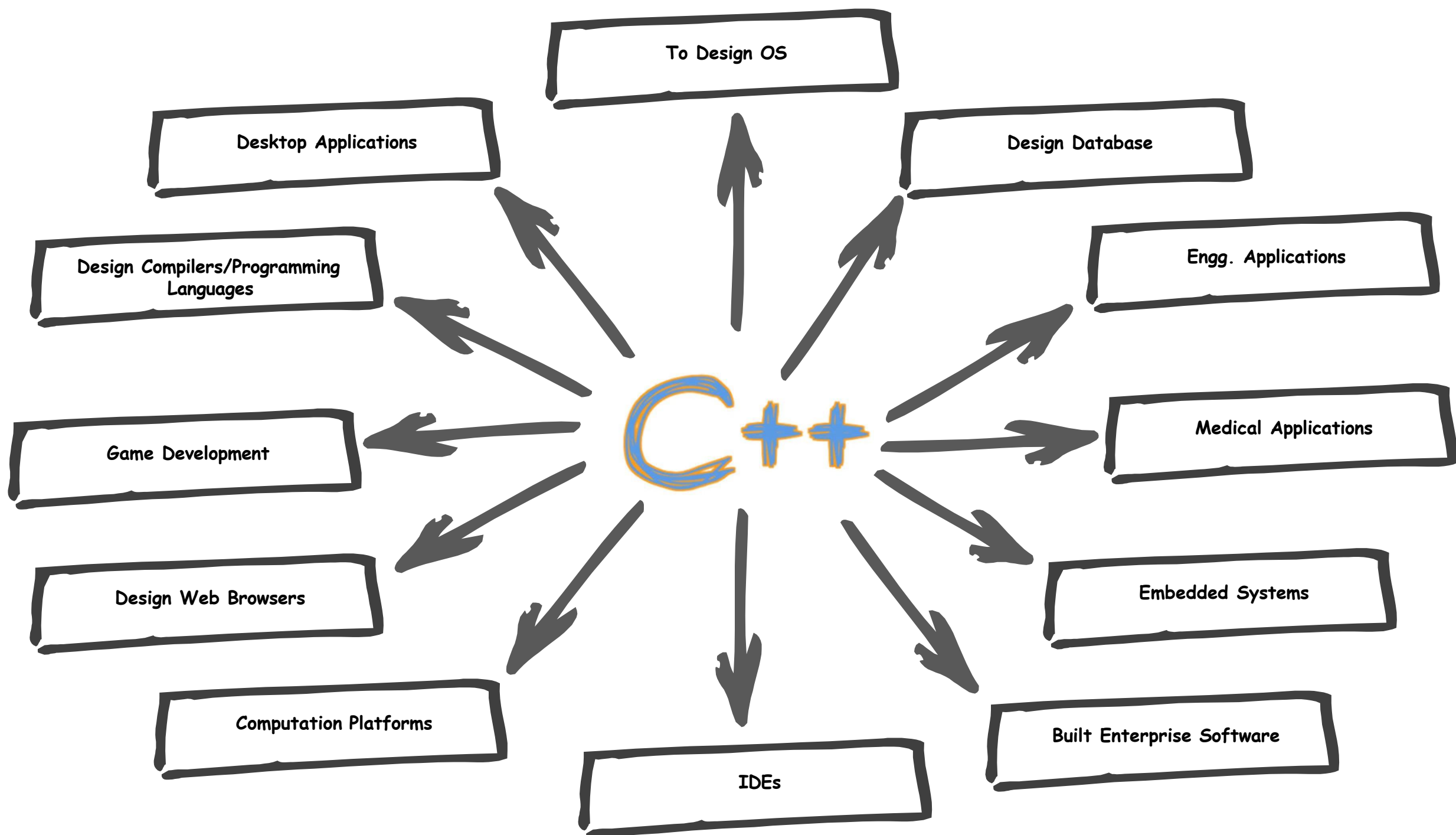
Libraries: chosen for **high-level math & performance; multithreading & concurrency** support.

Banking: concurrency & performance (e.g., Finacle backend).

Advanced Computation & Graphics: e.g., **Maya 3D** (animation, VR, 3D).

Cloud/Distributed: close to hardware + threading → great fit; **Bloomberg** stack note (C for DB engine; **C++** for dev environment/libraries).

When you talk about... performance, speed, cross-platform adaptability, heavy math, **multithreading/concurrency** → it's **C++**.





Programming Foundations

Level 1

Real Things done using C++

Operating Systems:



Most of the software from Microsoft is developed using C++ (flavors of Visual C++).
Applications like **Windows 95, ME, 98; XP**, etc. are written in C++.



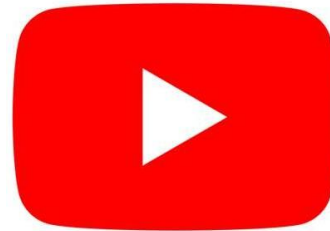
Apple OS X has some of its parts written in C++. Similarly, some parts of the iPod are also written in C++.

Applications:

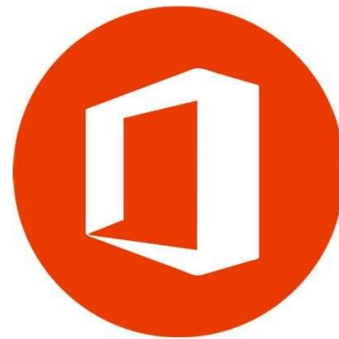


Adobe

Most of the applications of the adobe systems including **Illustrator**, **Photoshop**, etc.



Core Functions are written in C++



Microsoft products like **Word**, **Excel**, **PowerPoint**...etc



Amazon was primarily written in C++. Now C++ is mainly responsible for processing requests.

Browsers:

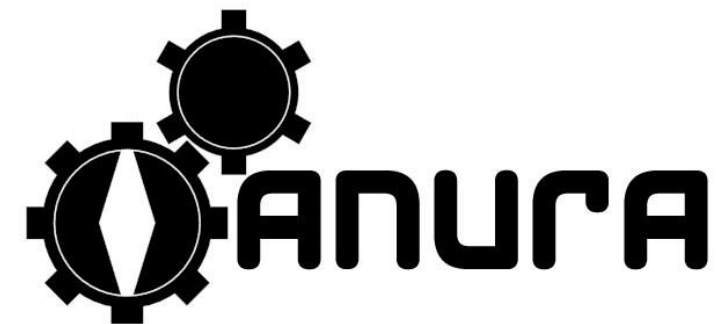


Database:



Gaming Engines:

A game engine is a dedicated software or an architecture, or program that developers use for creating games

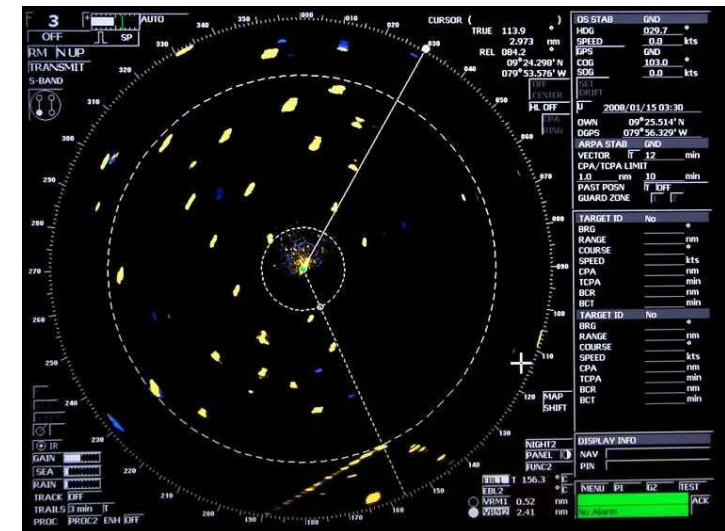


IDEs:



Enterprise Software

C++ is used in developing many enterprise software as well as advanced applications like **flight simulation** and **radar processing**.



Compilers and Programming Languages

Compilers of various high-level programming languages are written either in C or C++.

Many languages are written either



mingAdvices.com
Copyright © 2022

Embedded Systems

CARs, Flights, Medical Machines, and others

Libraries

When we require very high-level mathematical computations, performance and speed become important. Hence most of the libraries use C++ as their core programming language. Most **high-level machine language libraries use C++** as backend

C++ is faster than most of the other programming languages and also supports multithreading with concurrency.

Banking Applications

As C++ aids in concurrency, **it becomes the default choice for banking applications** that require multi-threading, concurrency, and high performance.



Infosys Finacle - is a popular core banking application that uses C++ as backend programming languages.

Advanced Computation And Graphics

C++ is useful in developing an application that requires **high-performance image processing, real-time physical simulations**, and **mobile sensor applications** that need high performance and speed.



Maya 3D software from Alias system is developed in C++ and is used for **animation, virtual reality, 3D graphics**, and environments.

Cloud/Distributed Systems

Cloud storage systems that are extensively used nowadays work close to the hardware. C++ becomes a default choice for implementing such systems as it is close to the hardware. C++ also provides multithreading support that can build concurrent applications and load tolerance.

The Bloomberg logo, consisting of the word "Bloomberg" in a bold, white, sans-serif font, centered within a solid black rectangular background.

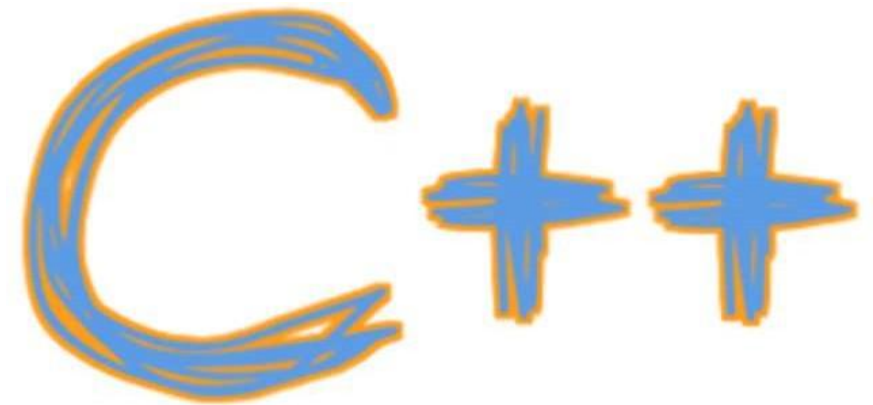
Bloomberg is a distributed RDBMS application that is used for accurately providing real-time financial information and news to investors.

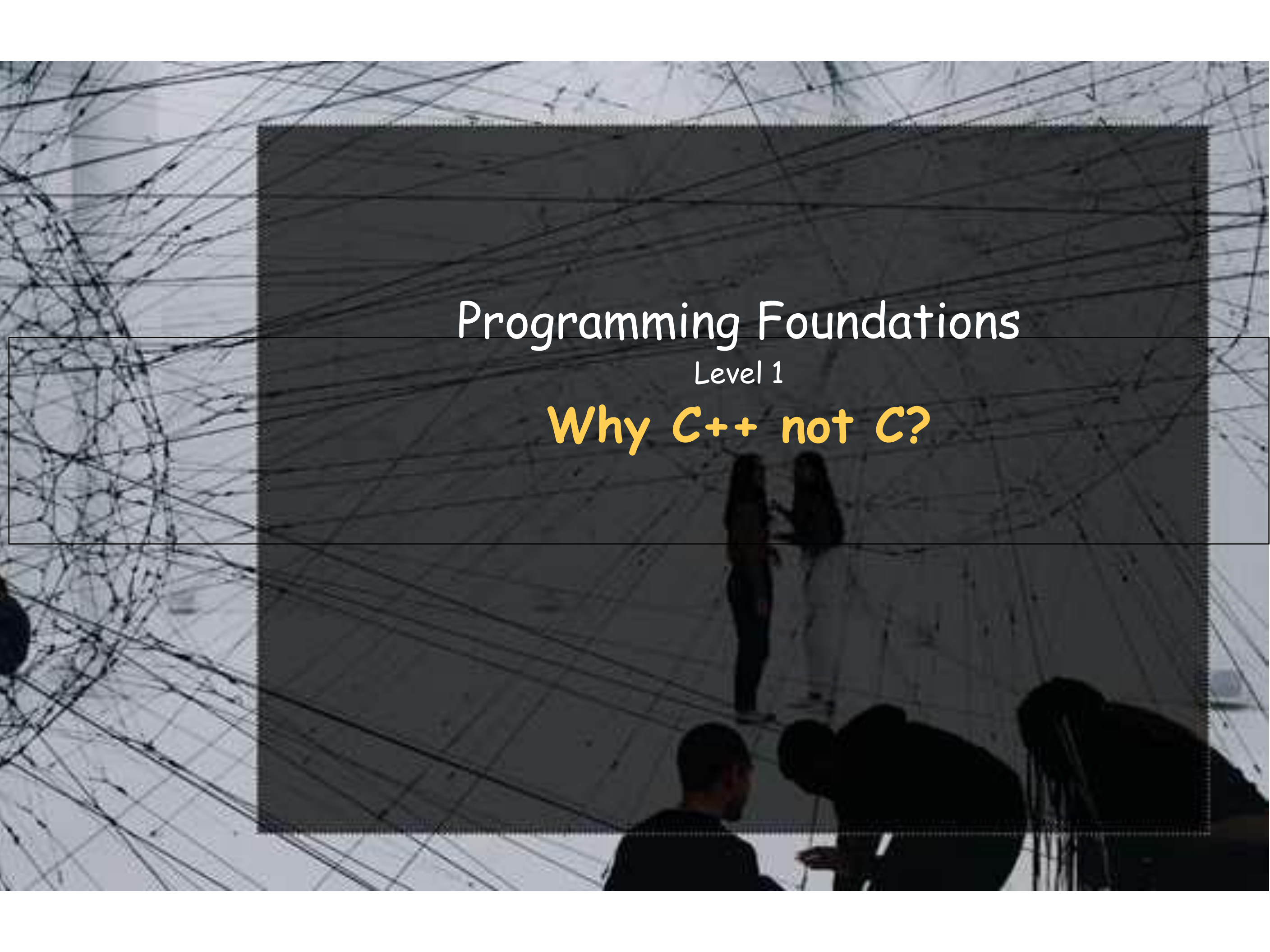
While Bloomberg's RDBMS is written in C, its development environment and set of libraries are written in C++.

Therefore, when you talk about...

- Performance
- Speed
- Cross Platform/Adaptable
- High-level mathematical computations
- Multithreading with concurrency

IT's...



The background of the slide features a complex, abstract network of thin, dark lines crisscrossing a light gray background. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a lecture hall. A large, dark gray rectangular area is centered on the slide, serving as a backdrop for the text.

Programming Foundations

Level 1

Why C++ not C?

C vs C++

Similarities

- Both the languages have a **similar Syntax/Grammar/Code Structure**.
- The **compilation** of both the languages **is similar**.
- Both are **Mid-level languages that are close to hardware** and are able to program and manipulate the underlying hardware resources.

Differences..

C

C was developed by Dennis Ritchie between the year 1969 and 1973 at AT&T Bell Labs.
C is a subset of C++.
Procedural Language.

C++

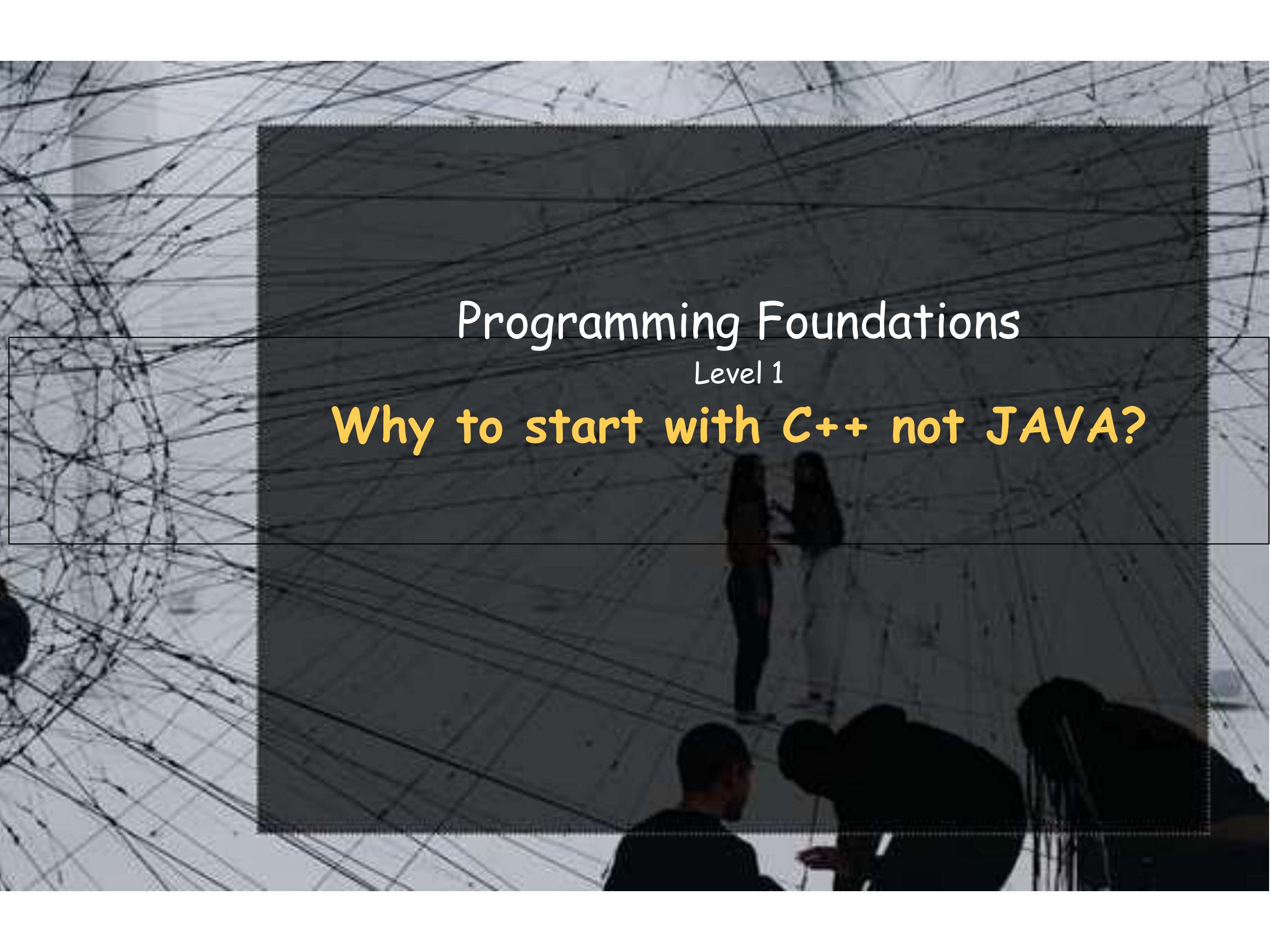
C++ was developed by Bjarne Stroustrup in 1979.
C++ Superset of C.
Supports both Procedural and OOP.

That's why you should learn C++ not C, it will take the same efforts

Similarities: syntax/grammar/compilation; both mid-level and close to hardware.

Differences: C is procedural, C++ is a superset that supports both procedural + OOP.

Conclusion : Learn C++ (not C) — same effort, broader power.

The background of the slide features a complex, dark network of thin, intersecting lines, resembling a web or a circuit board. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. The overall color palette is dark and monochromatic, with the text providing the primary points of contrast.

Programming Foundations

Level 1

Why to start with C++ not JAVA?

C++ vs Java

Similarities

- Both the great languages.

However, its recommended to start with C++ because:

C++

Supports both Procedural and OOP.

Java

Supports Only OOP.

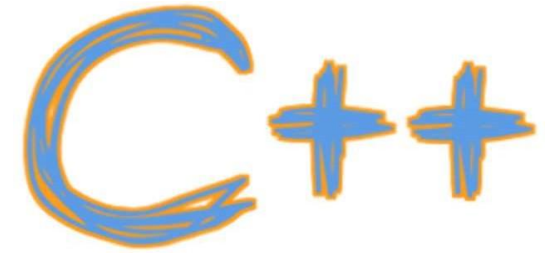
That's why you should start with C++, you will learn all types of programming.

C++ supports procedural + OOP, while Java is OOP-only.

Both are great; slides recommend starting with C++ so you learn all types of programming.

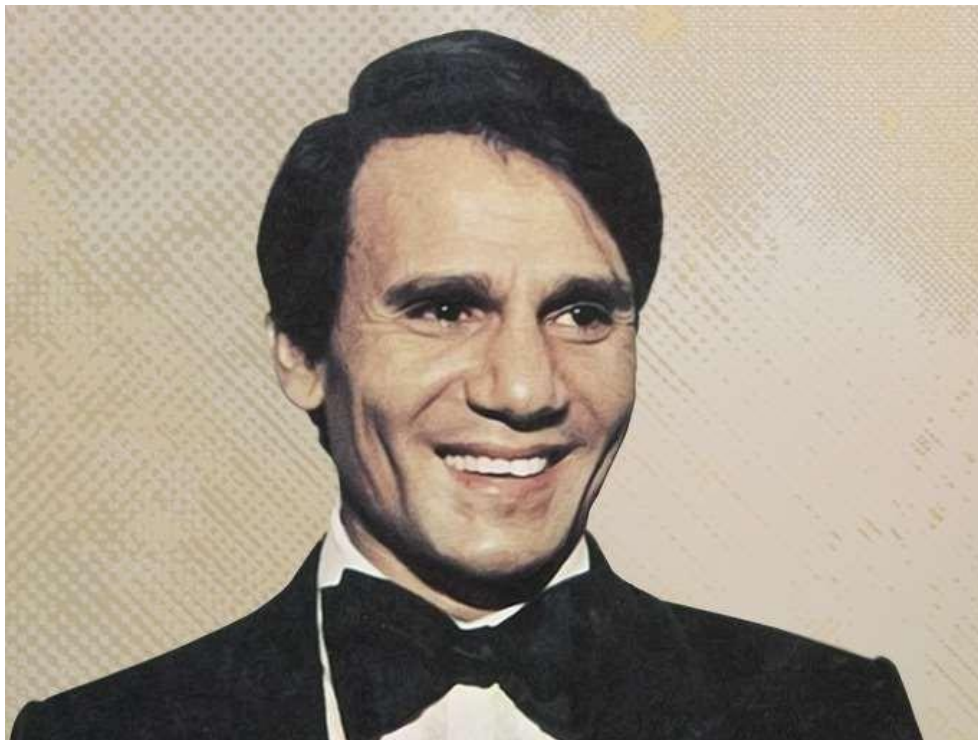
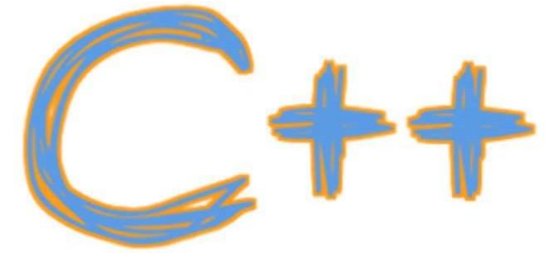
Will I work in C++?

- Yes and No.



Is it outdated and not used

- No, it still widely used for high performance systems.



And remember !!!!!

IF you learn C++, you can learn any other programming language easily in almost not time.



Python Developer
Learning C++

C++ Developer learning
Python





"Yes and No" (depends on role).

Not outdated — widely used in high-performance systems.

Even if you don't work using C++ , it will give you the keys to all programming languages and programming foundations.

C++ = Salary ++ even if you don't work using it!

Interconnection

- ◆ **C++ = mid-level control** → performance, memory control, concurrency
- ◆ **Paradigm coverage (procedural + OOP)** → broader mental model than single-paradigm starts.
- ◆ **Industry uses** (OS, browsers, engines, finance, graphics, cloud) → long-term relevance.

By starting with C++, you build **transferable fundamentals** that make every other language easier. 🚀

Activity

Instructions:

- 1 Give three reasons to start with C++.
- 2 Name five domains where C++ is used .
- 3 What paradigms does C++ support, and why does that matter when compared to Java?
- 4 Why do the slides say "Learn C++ not C" for beginners?
- 5 Explain two performance reasons C++ is chosen for infrastructure (libraries / cloud / banking).
- 6 True or False: "C++ is outdated."
Explain your answer.



Activity Solution



Solution:

- 1 Give three reasons to start with C++.
 - Cross-platform and high-performance.
 - Teaches both procedural + OOP programming styles.
 - Builds core fundamentals, making other languages easy to learn.
- 2 Name five domains where C++ is used .
 - Operating systems
 - Web browsers
 - Game engines
 - Enterprise / banking systems
 - Embedded & medical systems
 - Databases, IDEs, graphics tools (e.g., Maya), and cloud computing.
- 3 What paradigms does C++ support, and why does that matter when compared to Java?
 - C++ supports Procedural + Object-Oriented Programming (OOP).
 - Java is OOP-only, so C++ provides a broader foundation for beginners.
- 4 Why "Learn C++ not C" for beginners?
 - Same learning effort, but C++ is a superset of C.
 - Covers more paradigms and teaches stronger concepts.

- 5 Explain two performance reasons C++ is chosen for infrastructure (libraries / cloud / banking).

Closer to hardware for faster execution and better memory control.

Supports multithreading and concurrency for scalable, high-performance systems.

- 6 True or False: "C++ is outdated."

Explain your answer.

✗ False.

C++ is still widely used in performance-critical systems (OS, finance, cloud, graphics, and gaming).

Conclusion

- ✓ Start with **C++** to gain **deep, transferable fundamentals**.
- ✓ C++ powers **OS, browsers, engines, finance, embedded, graphics, cloud** — proving long-term value.
- ✓ With C++ as your base, **other languages come quickly**, and you can choose the **best tool for each job**.

Thank you 😊